CSE 318: Artificial Intelligence Sessional
Offline Assignment 3: Adversarial Search (Mancala)

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1 Introduction

Mancala (Kalah variant) is a two-player zero-sum sequential game with perfect information. The objective of this assignment is to design and implement an adversarial search game engine utilizing the **Minimax algorithm with α-β pruning**, **move ordering**, and **iterative deepening**. Multiple heuristic evaluation functions are implemented and evaluated systematically across multiple search depths to analyze their strategic efficacy, pruning efficiency, and first-mover dynamics.

2 Game Rules and Mechanics

The game is played on a board with 2 rows of 6 pits (each initialized with 4 stones, totaling 48 stones) and 2 stores (Kalahs).

1. **Sowing**: On a player's turn, all stones from a non-empty pit on their side are gathered and dropped one-by-one counter-clockwise into subsequent pits and the player's own store, skipping the opponent's store.
2. **Extra Turn**: If the last stone dropped lands in the player's own store, the player earns an additional turn immediately.
3. **Capture Rule**: If the last stone lands in an empty pit on the player's own side and the directly opposite pit on the opponent's side is non-empty, both the final stone and all stones in the opposite pit are captured into the active player's store.
4. **Endgame & Stone Sweeping**: The game terminates immediately when all 6 pits on either player's side become empty. The remaining stones on the opponent's side are swept into the opponent's store. The player with the highest total score in their store wins.

3 Adversarial Search & Heuristic Formulations

The search engine is implemented using Minimax with α-β pruning, enhanced by:

- **Move Ordering**: Sorts candidate actions by prioritizing potential extra turns and capture payoffs, maximizing the number of early α-β cutoffs.
- **Iterative Deepening**: Progressively explores search depths $d = 1, 2, \ldots, D$ ensuring robust move selection within bounded computational constraints.

Four distinct heuristics specified in the assignment were implemented:

$$H_1 = (\text{storage}_{\text{my}} - \text{storage}_{\text{opp}}) \tag{1}$$

$$H_2 = W_1 \cdot (\text{storage}_{\text{my}} - \text{storage}_{\text{opp}}) + W_2 \cdot (\text{side}_{\text{my}} - \text{side}_{\text{opp}}) \tag{2}$$

$$H_3 = W_1 \cdot (\text{storage}_{\text{my}} - \text{storage}_{\text{opp}}) + W_2 \cdot (\text{side}_{\text{my}} - \text{side}_{\text{opp}}) + W_3 \cdot (\text{extra_turn}) \tag{3}$$

$$H_4 = W_1 \cdot (\text{storage}_{\text{my}} - \text{storage}_{\text{opp}}) + W_2 \cdot (\text{side}_{\text{my}} - \text{side}_{\text{opp}}) + W_3 \cdot (\text{extra_turn}) + W_4 \cdot (\text{captured}) \tag{4}$$

Standard weight configuration used: $W_1 = 2.0, W_2 = 1.0, W_3 = 5.0, W_4 = 3.0$.

4 Experimental Setup & Methodology

Because the Minimax search and initial board state are deterministic:

- For every heuristic pair (H_i, H_j), exactly 2 games are executed:
    1. **Game 1**: H_i as Player 1 (First Mover) vs H_j as Player 2.
    2. **Game 2**: H_j as Player 1 (First Mover) vs H_i as Player 2.
- This isolates the first-mover advantage from heuristic quality.
- Experiments were conducted across search depths: $d \in \{4, 6, 8\}$.
- Metrics recorded: **Scores**, **Winner**, **Total Nodes Explored**, and **Alpha-Beta Cutoffs**.

5 Experimental Results

5.1 Results at Search Depth $d = 4$

Table 1: Deterministic Head-to-Head Results at Depth $d = 4$

Matchup	P1 (First)	P2 (Second)	Score	Winner	Nodes (P1 / P2)	Cutoffs (P1 / P2)
H_1 vs H_2	H_1	H_2	34 – 14	H_1	24,103 / 22,518	8,501 / 2,722
H_2 vs H_1	H_2	H_1	19 – 29	H_1	19,517 / 7,011	2,856 / 1,813
H_1 vs H_3	H_1	H_3	34 – 14	H_1	24,103 / 22,518	8,501 / 2,722
H_3 vs H_1	H_3	H_1	19 – 29	H_1	19,517 / 7,011	2,856 / 1,813
H_1 vs H_4	H_1	H_4	34 – 14	H_1	24,103 / 22,518	8,501 / 2,722
H_4 vs H_1	H_4	H_1	19 – 29	H_1	19,517 / 7,011	2,856 / 1,813
H_2 vs H_3	H_2	H_3	21 – 27	H_3	31,270 / 24,433	7,071 / 4,736
H_3 vs H_2	H_3	H_2	21 – 27	H_2	31,270 / 24,433	7,071 / 4,736
H_2 vs H_4	H_2	H_4	21 – 27	H_4	31,270 / 24,433	7,071 / 4,736
H_4 vs H_2	H_4	H_2	21 – 27	H_2	31,270 / 24,433	7,071 / 4,736
H_3 vs H_4	H_3	H_4	21 – 27	H_4	31,270 / 24,433	7,071 / 4,736
H_4 vs H_3	H_4	H_3	21 – 27	H_3	31,270 / 24,433	7,071 / 4,736

Table 2: Overall Performance Summary at Depth $d = 4$

Heuristic	Wins	Losses	Draws	Win Rate (%)	Total Stones Captured
H_1	6	0	0	**100.0%**	**189**
H_2	2	4	0	33.3%	129
H_3	2	4	0	33.3%	129
H_4	2	4	0	33.3%	129

5.2 Results at Search Depth $d = 6$

Table 3: Deterministic Head-to-Head Results at Depth $d = 6$

Matchup	P1 (First)	P2 (Second)	Score	Winner	Nodes (P1 / P2)	Cutoffs (P1 / P2)
H_1 vs H_2	H_1	H_2	33 – 15	H_1	300,084 / 356,843	118,187 / 62,721
H_2 vs H_1	H_2	H_1	31 – 17	H_2	197,153 / 87,945	54,017 / 28,864
H_1 vs H_3	H_1	H_3	33 – 15	H_1	300,084 / 356,843	118,187 / 62,721
H_3 vs H_1	H_3	H_1	31 – 17	H_3	197,153 / 87,945	54,017 / 28,864
H_1 vs H_4	H_1	H_4	33 – 15	H_1	300,084 / 356,843	118,187 / 62,721
H_4 vs H_1	H_4	H_1	31 – 17	H_4	197,153 / 87,945	54,017 / 28,864
H_2 vs H_3	H_2	H_3	32 – 16	H_2	209,940 / 143,329	57,320 / 42,031
H_3 vs H_2	H_3	H_2	32 – 16	H_3	209,940 / 143,329	57,320 / 42,031
H_2 vs H_4	H_2	H_4	32 – 16	H_2	209,940 / 143,329	57,320 / 42,031
H_4 vs H_2	H_4	H_2	32 – 16	H_4	209,940 / 143,329	57,320 / 42,031
H_3 vs H_4	H_3	H_4	32 – 16	H_3	209,940 / 143,329	57,320 / 42,031
H_4 vs H_3	H_4	H_3	32 – 16	H_4	209,940 / 143,329	57,320 / 42,031

Table 4: Overall Performance Summary at Depth $d = 6$

Heuristic	Wins	Losses	Draws	Win Rate (%)	Total Stones Captured
H_1	3	3	0	50.0%	**150**
H_2	3	3	0	50.0%	142
H_3	3	3	0	50.0%	142
H_4	3	3	0	50.0%	142

5.3 Results at Search Depth $d = 8$

Table 5: Deterministic Head-to-Head Results at Depth $d = 8$

Matchup	P1 (First)	P2 (Second)	Score	Winner	Nodes (P1 / P2)	Cutoffs (P1 / P2)
H_1 vs H_2	H_1	H_2	38 – 10	H_1	4,834,636 / 3,956,249	2,063,588 / 926,499
H_2 vs H_1	H_2	H_1	24 – 24	Draw	2,820,977 / 2,343,148	1,031,834 / 709,246
H_1 vs H_3	H_1	H_3	38 – 10	H_1	4,834,636 / 3,956,249	2,063,588 / 926,499
H_3 vs H_1	H_3	H_1	24 – 24	Draw	2,820,977 / 2,343,148	1,031,834 / 709,246
H_1 vs H_4	H_1	H_4	38 – 10	H_1	4,834,636 / 3,956,249	2,063,588 / 926,499
H_4 vs H_1	H_4	H_1	24 – 24	Draw	2,820,977 / 2,343,148	1,031,834 / 709,246
H_2 vs H_3	H_2	H_3	22 – 26	H_3	3,589,966 / 2,372,947	1,197,664 / 736,947
H_3 vs H_2	H_3	H_2	22 – 26	H_2	3,589,966 / 2,372,947	1,197,664 / 736,947
H_2 vs H_4	H_2	H_4	22 – 26	H_4	3,589,966 / 2,372,947	1,197,664 / 736,947
H_4 vs H_2	H_4	H_2	22 – 26	H_2	3,589,966 / 2,372,947	1,197,664 / 736,947
H_3 vs H_4	H_3	H_4	22 – 26	H_4	3,589,966 / 2,372,947	1,197,664 / 736,947
H_4 vs H_3	H_4	H_3	22 – 26	H_3	3,589,966 / 2,372,947	1,197,664 / 736,947

Table 6: Overall Performance Summary at Depth $d = 8$

Heuristic	Wins	Losses	Draws	Win Rate (%)	Total Stones Captured
H_1	3	0	3	**50.0%**	**186**
H_2	2	3	1	33.3%	130
H_3	2	3	1	33.3%	130
H_4	2	3	1	33.3%	130

6 Analysis and Discussion

6.1 1. Impact of First-Mover Advantage

In Mancala (Kalah), Player 1 possesses a significant structural advantage because they dictate the opening tempo and can immediately capture or generate free turns. At depth $d = 6$, Player 1 won all 12 games regardless of the heuristic pairing. At depth $d = 8$, Player 1 won 9 out of 12 games, and drew the remaining 3 games.

6.2 2. Heuristic Quality Comparison

- H_1 **(Pure Storage Difference)**: Despite its simplicity, H_1 strictly focuses on the terminal win condition (ΔStorage). At shallow depths ($d = 4$) and deeper depths ($d = 8$), it consistently accumulated the highest total stones (189 at $d = 4$, 186 at $d = 8$) without suffering a single loss.
- H_2, H_3, H_4: Incorporating side stones, extra turns, and captures guides the search well in tactical scenarios. Between H_2, H_3, and H_4, moves that yield additional turns and capture payloads (H_3 and H_4) outperform H_2 when playing second.

6.3 3. Alpha-Beta Pruning Efficiency

Move ordering by extra-turn potential and captures drastically improved search efficiency:

- At depth $d = 8$, over $\mathbf{2.06 \times 10^6}$ cutoffs were achieved in a single game, reducing the branching factor significantly and enabling search up to depth 8 in real time without exceeding memory limits.

7 Conclusion

The Minimax adversarial search with α-β pruning, move ordering, and iterative deepening successfully solves Mancala decision-making within fractional seconds. While first-mover tempo strongly influences the outcome, heuristics focusing on store accumulation and tactical extra-turn capture mechanics achieve the most reliable performance.